Lab 3,4 and 5:-

Q- WRITE A C PROGRAM TO FIND THE VALUE OF Y USING

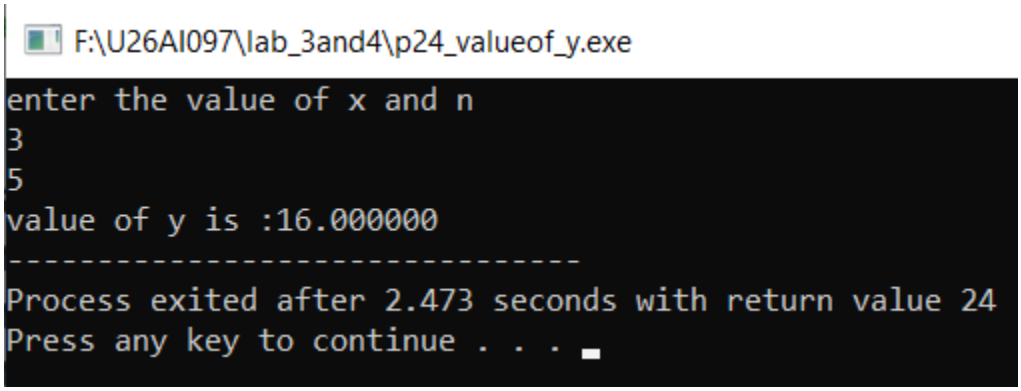
$$y = \begin{cases} 1+x & \text{where } n=1 \\ 1+x/n & \text{where } n=2 \\ 1+xn & \text{when } n=3 \\ 1+nx & \text{when } n>3 \text{ or } n<1 \end{cases}$$

Code:-

```
#include<stdio.h>
main()
{
	float x,n,y;
	printf("enter the value of x and n\n");
	scanf("%f%f",&x,&n);
	if(n==1)
	{
		y=1+x;
		printf("value of y is: %f",y);
	}
	else if(n==2)
	{
		y=1+(x*0.5);
		printf("value of y is: %f",y);
	}
	else if(n==3)
	{
		y=1+(x*x*x);
		printf("value of y is: %f",y);
	}
	else
	{
```

```
            y=1+(n*x);
            printf("value of y is :%f",y);
        }
}
```

Output:

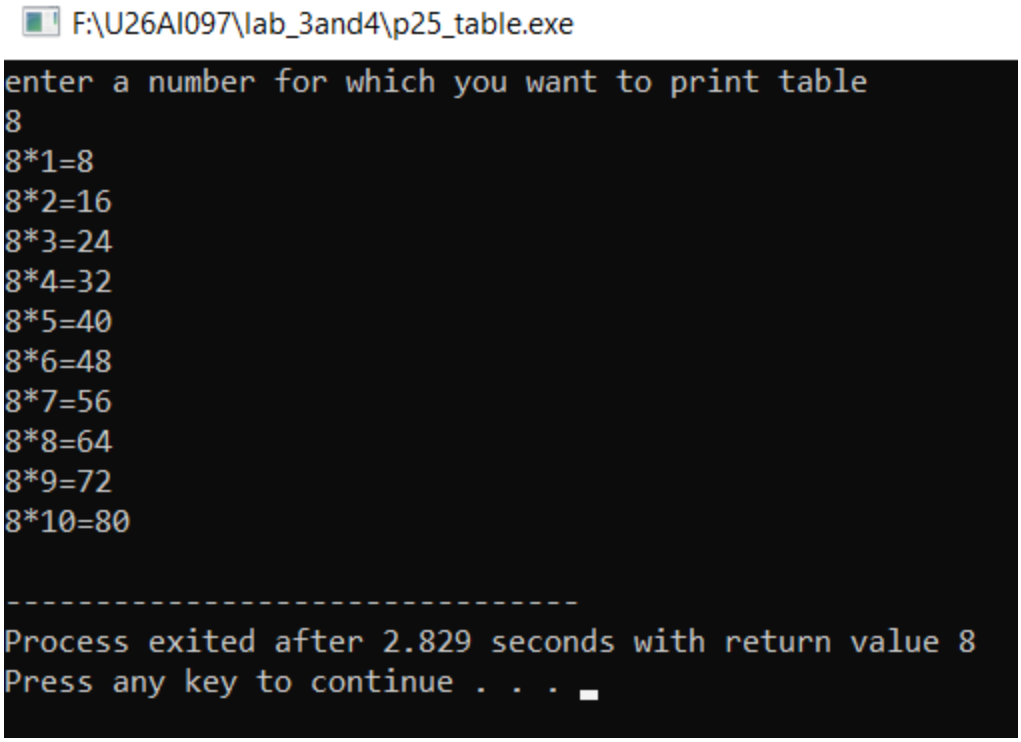


Q-25: PROGRAM TO PRINT TABLE OF ANY NO.

Code:

```
#include<stdio.h>
int main()
{
        int number,i,x;
        printf("enter a number for which you want to print table\n");
        scanf("%d",&number);
        for(i=1;i<=10;i++)
        {
            x=number*i;
            printf("%d*%d=%d\n",number,i,x);
        }
}
```

Output:



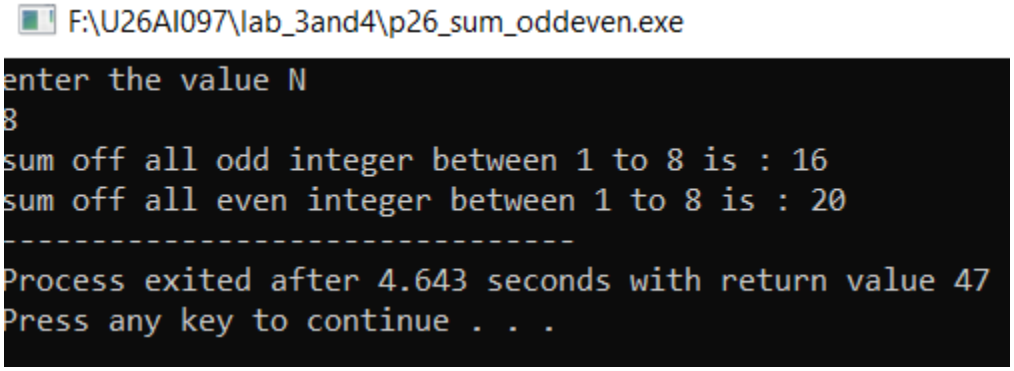
Q:26. PROGRAM TO PRINT SUM OF ALL ODD AND EVEN NO B/W 1 TO N.

Code:

```
#include<stdio.h>
main()
{
	int i,N,sum_odd=0,sum_even=0;
	printf("enter the value N\n");
	scanf("%d",&N);
	for(i=1;i<=N;i++)
	{
		if(i%2!=0)
		{
		sum_odd=sum_odd+i;
```

```
            }
        }
        printf("sum off all odd integer between 1 to %d is :
%d\n",N,sum_odd);
        for(i=0;i<=N;i++)
        {
            if(i%2==0)
            {
            sum_even=sum_even+i;
            }
        }
        printf("sum off all even integer between 1 to %d is :
%d",N,sum_even);
}
```

Output:

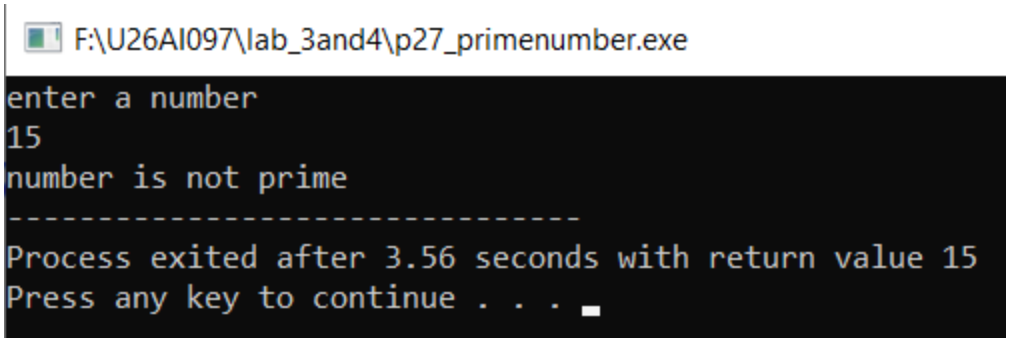


Q-27. PROGRAM TO CHECK WHETHER A NO IS PRIME OR NOT.

Code:

```
#include<stdio.h>
int main()
{
	int num,i,flag=0;
	printf("enter a number\n");
	scanf("%d",&num);
	for(i=2;i<num;i++)
	{
		if(num%i==0 && num!=1)
		{
			flag=1;
	 }
	}
	if(flag=1)
	printf("number is not prime");
	else
	printf("number is prime");
	if(num==1)
	printf("it is a unique number",num);
	else if(num==2)
	printf("it is a prime number",num);
}
```

Output:

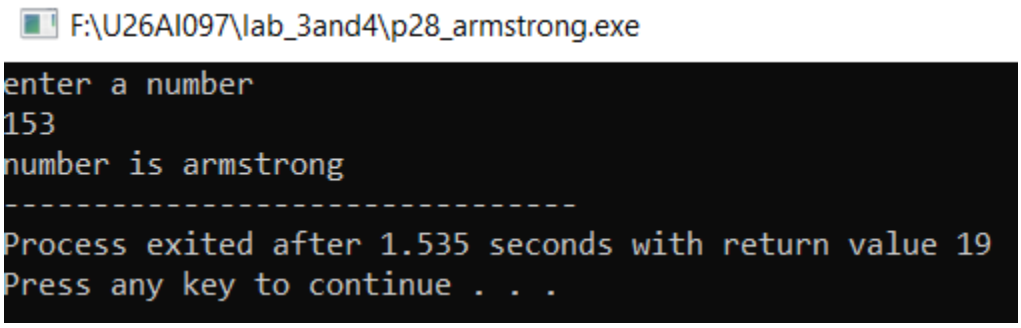


Q:28. PROGRAM TO CHECK WHETHER A NO IS ARMSTRONG OR NOT.

Code:

```
#include<stdio.h>
int main()
{
	int num,q,sum,tamp;
	sum=0;
	printf("enter a number\n");
	scanf("%d",&num);
	tamp=num;
	while(num!=0)
	{
		q=num%10;
		sum=sum+q*q*q;
		num=num/10;
	}
	if(tamp==sum)
	printf("number is armstrong");
	else
	printf("number is not armstrong");
}
```

Output:



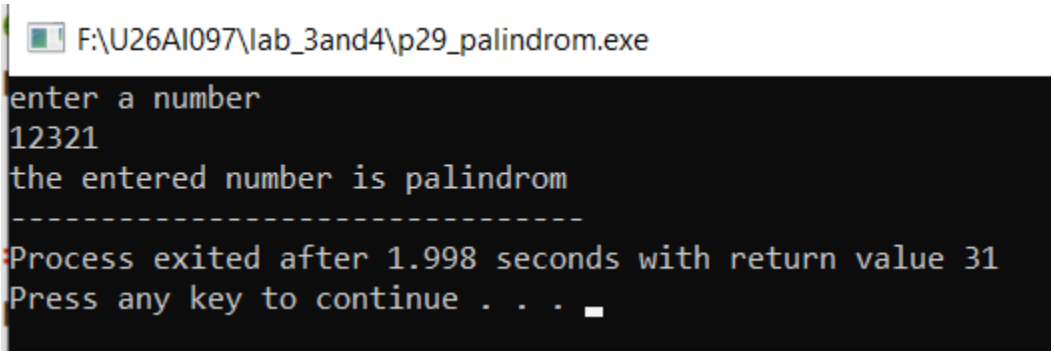
Q:29. PROGRAM TO CHECK WHETHER A NO IS PALINDROM OR NOT.

CODE:

```
#include<stdio.h>
int main()
{
	int num,q,sum,tamp;
	sum=0;
	printf("enter a number\n");
	scanf("%d",&num);
	tamp=num;
	while(num!=0)
	{
		q=num%10;
		sum=sum*10+q;
		num=num/10;
	}
	if(tamp==sum)
	printf("the entered number is palindrom",tamp);
```

```
        else
        printf("the entered number is not palindrom",tamp);
}
```

OUTPUT:

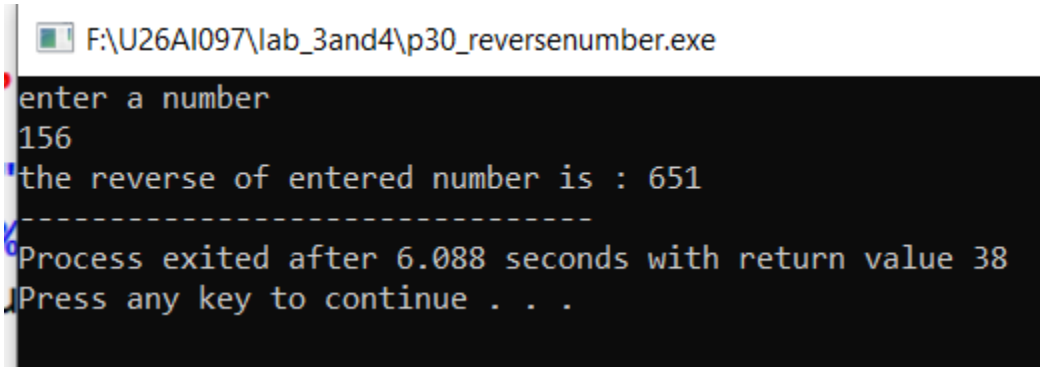


Q:30. PROGRAM TO REVERSE OF A GIVEN NO.

CODE:

```
#include<stdio.h>
int main()
{
        int num,q,sum;
        sum=0;
        printf("enter a number\n");
        scanf("%d",&num);
        while(num!=0)
        {
                q=num%10;
                sum=sum*10+q;
                num=num/10;
        }
        printf("the reverse of entered number is : %d",sum);
}
```

OUTPUT:

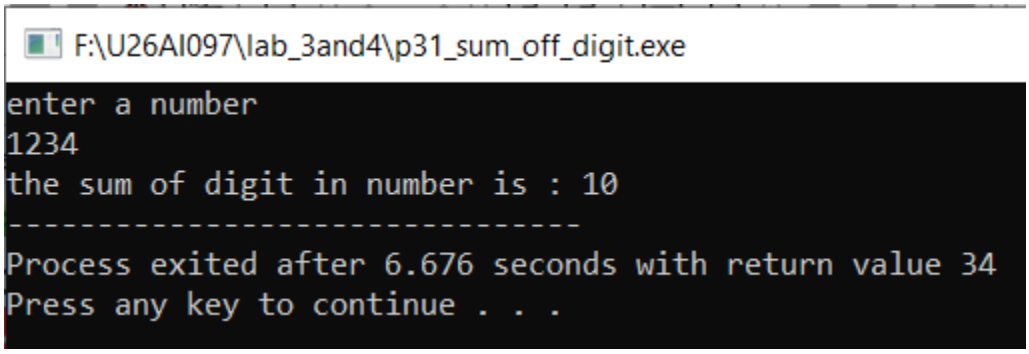


Q:31. PROGRAM TO PRINT SUM OF INDIVIDUAL DIGITS OF A ‘N’ DIGIT NO.

CODE:

```
#include<stdio.h>
int main()
{
	int num,q,sum;
	sum=0;
	printf("enter a number\n");
	scanf("%d",&num);
	while(num!=0)
	{
		q=num%10;
		sum=sum+q;
		num=num/10;
	}
	printf("the sum of digit in number is : %d",sum);
}
```

OUTPUT:



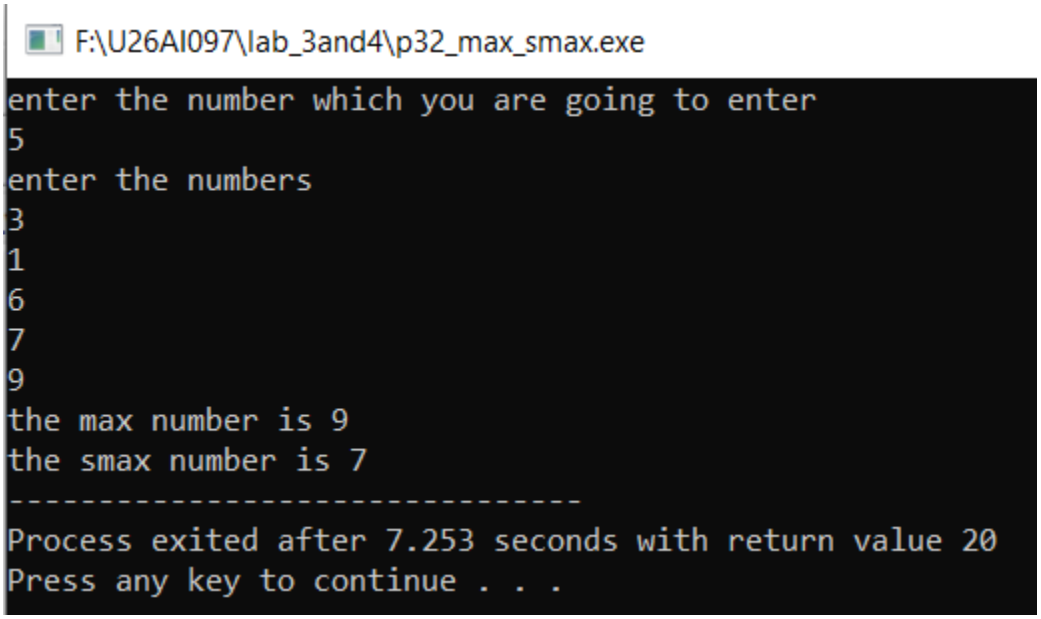
Q:32. PROGRAM TO FINDOUT MAX AND SECOND MAX FROM ‘N’ NOS .

CODE:

```
#include<stdio.h>
#include<limits.h>
int main()
{
	int n,i;
	printf("enter the number which you are going to enter\n");
	scanf("%d",&n);
	printf("enter the numbers\n");
	int number[n];
	for(i=0;i<n;i++)
	{
		scanf("%d",&number[i]);
	}
	int max=INT_MIN;
	for(i=0;i<n;i++)
	{
		if(max<number[i])
		max=number[i];
```

```
    }
    int smax=INT_MIN;
    for(i=0;i<n;i++)
    {
        if( number[i]!=max&&smax<number[i])
        smax=number[i];
    }
    printf("the max number is %d\n",max);
    printf("the smax number is %d",smax);
}
```

OUTPUT:



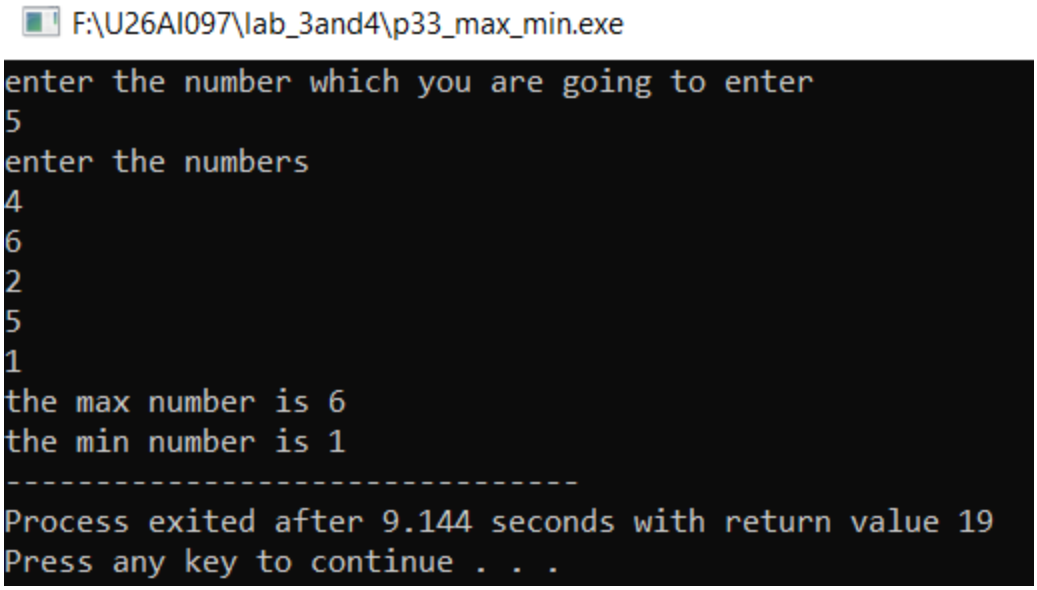
Q:33. PROGRAM TO FINDOUT MAX AND MIN FROM ‘N’ NOS.

CODE:

```
#include<stdio.h>
#include<limits.h>
```

```
int main()
{
	int n,i;
	printf("enter the number which you are going to enter\n");
	scanf("%d",&n);
	printf("enter the numbers\n");
	int number[n];
	for(i=0;i<n;i++)
	{
		scanf("%d",&number[i]);
	}
	int max=INT_MIN;
	for(i=0;i<n;i++)
	{
		if(max<number[i])
		max=number[i];
	}
	int min=number[0];
	for(i=0;i<n;i++)
	{
		if(min>number[i])
		min=number[i];
	}
	printf("the max number is %d\n",max);
	printf("the min number is %d",min);
}
```

OUTPUT:



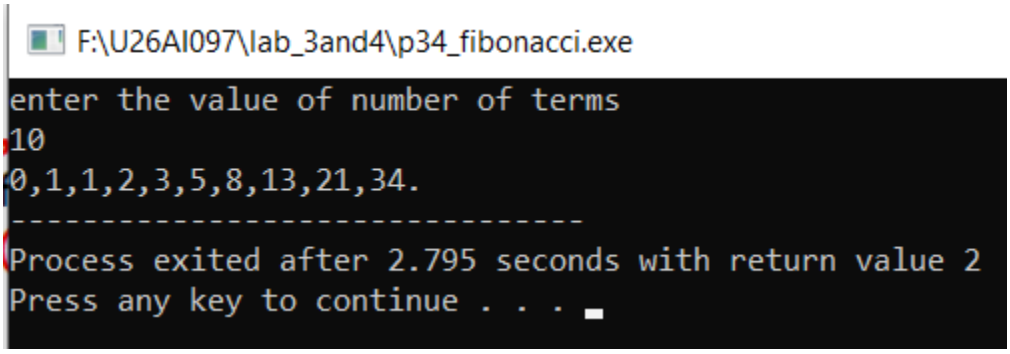
Q:34. PROGRAM TO PRINT FIBONACCI SERIES.(0,1,1,2,3…)

CODE:

```
#include<stdio.h>
main()
{
	int n,f,f1,f2,i;
	printf("enter the value of number of terms\n");
	scanf("%d",&n);
	f1=0;
	f2=1;
	printf("%d,%d,",f1,f2);
	for(i=1;i<=n-2;i++)
	{
		f=f1+f2;
		f1=f2;
		f2=f;
		printf("%d,",f);
```

```
        }
        printf("\b.",f);
}
```

OUTPUT:



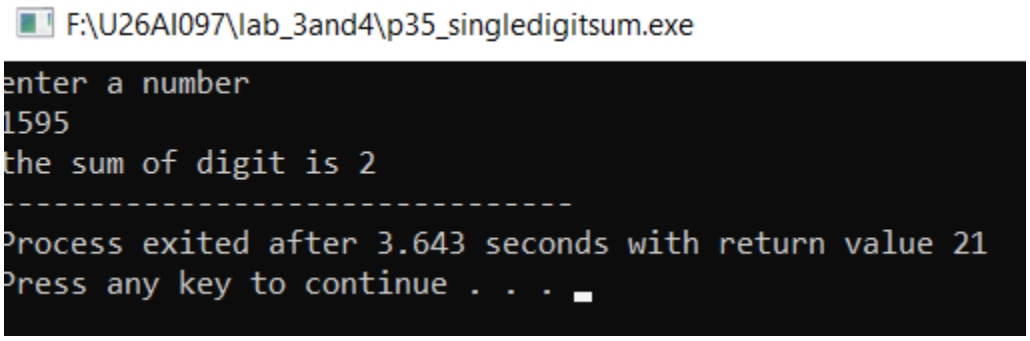
Q:35. READ A NUMBER N AND PRINT A SINGLE DIGIT ANSWER SHOWING SUM OF DIGIT OF N

CODE:

```
#include<stdio.h>
int main()
{
        int num,sum,q;
        printf("enter a number\n");
        scanf("%d",&num);
        while(num>=10)
        {
                sum=0;
                while(num>0)
                {
                q=num%10;
                sum=sum+q;
                num=num/10;
```

```
        }
        num=sum;
    }
    printf("the sum of digit is %d",num);
}
```

OUTPUT:

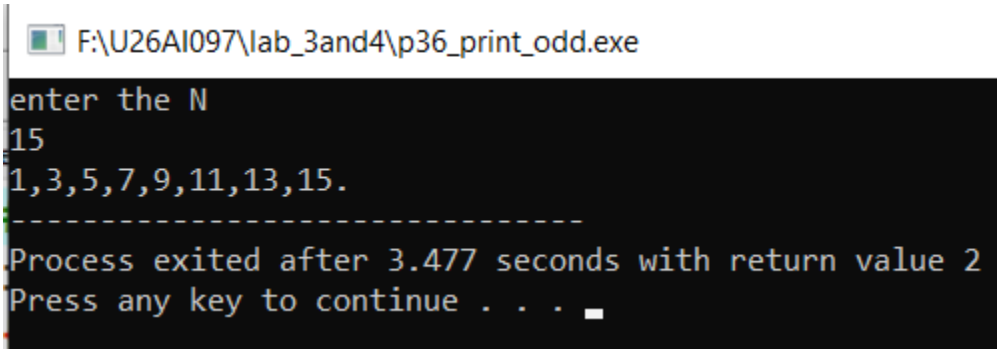


Q:36. PROGRAM TO PRINT 1,3,5,7,9……N.

CODE:

```
#include<stdio.h>
int main()
{
    int N,i;
    printf("enter the N\n");
    scanf("%d",&N);
    for(i=1;i<=N;i+=2)
    {
        printf("%d,",i);
    }
    printf("\b.",i);
}
```

OUTPUT:

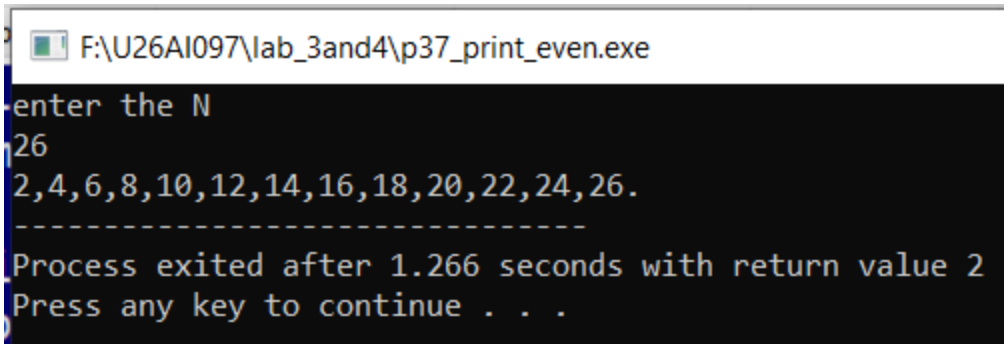


Q:37. PROGRAM TO PRINT 2,4,6,8,10,12……N.

CODE:

```
#include<stdio.h>
int main()
{
	int N,i;
	printf("enter the N\n");
	scanf("%d",&N);
	for(i=2;i<=N;i+=2)
	{
		printf("%d,",i);
	}
	printf("\b.",i);
}
```

OUTPUT:

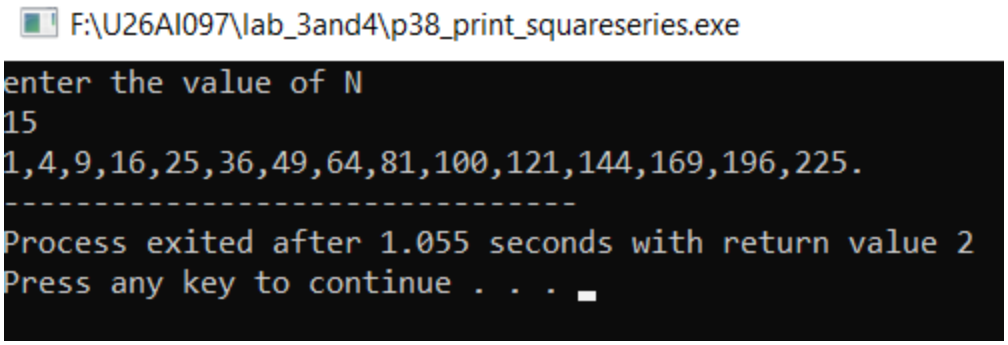


Q:38. PROGRAM TO PRINT 1,4,9,16,25,……N.

CODE:

```
#include<stdio.h>
int main()
{
	int N,i,x;
	printf("enter the value of N\n");
	scanf("%d",&N);
	for(i=1;i<=N;i++)
	{
		x=i*i;
		printf("%d,",x);
	}
	printf("\b.",x);
}
```

OUTPUT:

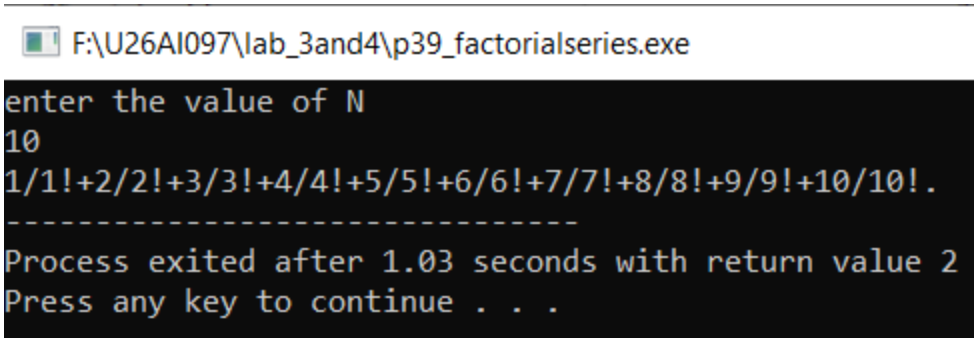


Q:39. PROGRAM TO PRINT 1/1!+2/2!+3/3!+4/4!,……N.

CODE:

```
#include<stdio.h>
int main()
{
	int N,i,x;
	printf("enter the value of N\n");
	scanf("%d",&N);
	for(i=1;i<=N;i++)
	{
		x=i;
		printf("%d/%d!+",i,i);
	}
	printf("\b.",i);
}
```

OUTPUT:



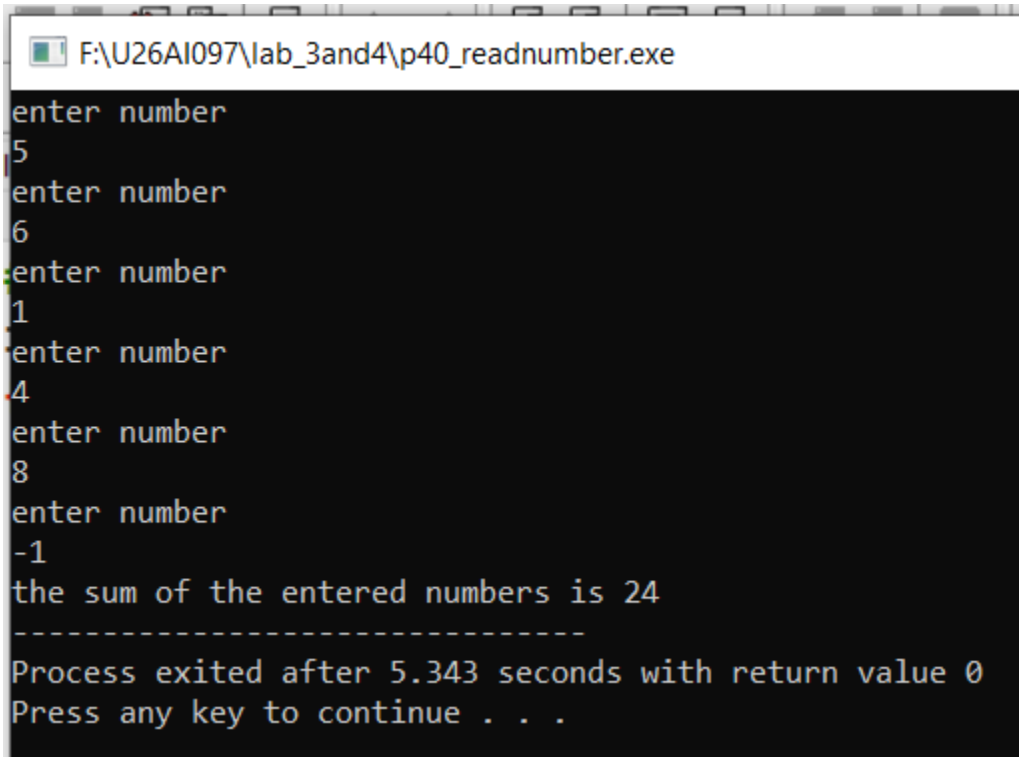
Q:40. READ NUMBERS TILL A NEGATIVE NUMBER IS ENTERED AND CALCULATE SUM OF A LIST OF NUMBERS READ.

CODE:

```
#include<stdio.h>
int main()
{
	int num,sum=0;
	printf("enter number\n");
	scanf("%d",&num);
	sum=num;
	if(sum>=0)
	{
	while(num>0)
	{
		printf("enter number\n");
		scanf("%d",&num);
		if(num<0)
		{
		break;
		}
```

```
            sum=sum+num;
        }
        printf("the sum of the entered numbers is %d",sum);
        }
        return 0;
}
```

OUTPUT:



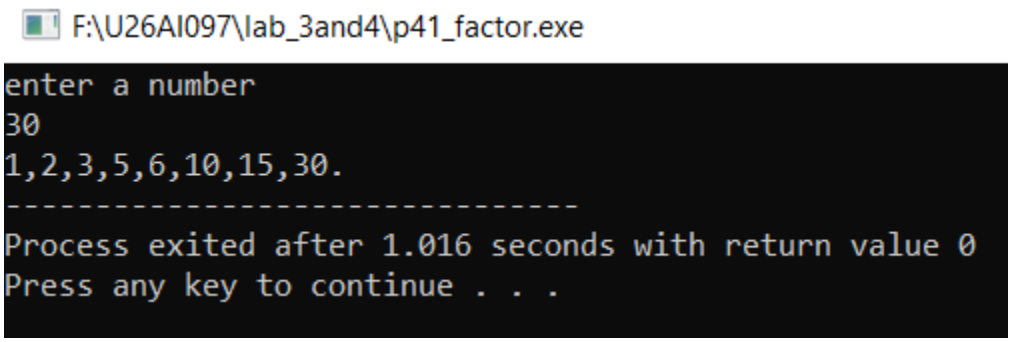
Q:41. READ A NUMBER N AND PRINT FACTOR OF N.(EX 28= 1,2,4,7,14,28)

CODE:

```
#include<stdio.h>
int main()
{
```

```
	int num,i;
	printf("enter a number\n");
	scanf("%d",&num);
	for(i=1;i<=num;i++)
	{
		if(num%i==0)
		{
			printf("%d,",i);
		}
	}
	printf("\b.");
	return 0;
}
```

OUTPUT:



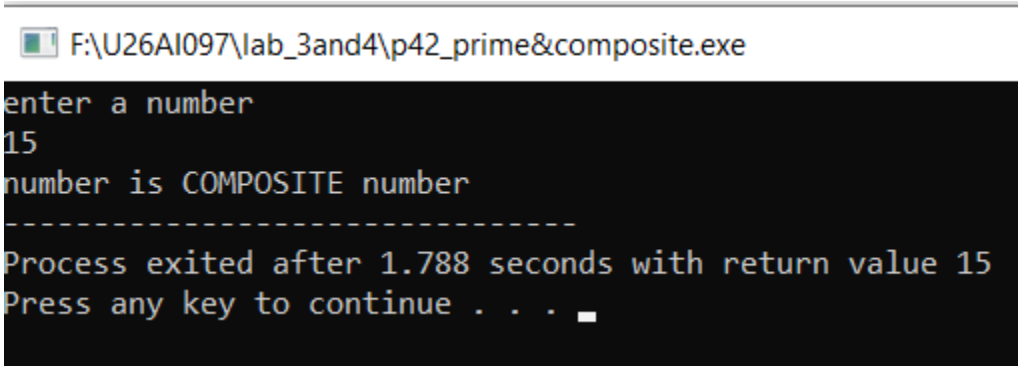
Q:42. READ A NUMBER N AND PRINT IF N IS PRIME OR COMPOSITE.

CODE:

```
#include<stdio.h>
int main()
{
	int num,i,flag=0;
```

```
        printf("enter a number\n");
        scanf("%d",&num);
        for(i=2;i<num;i++)
        {
                if(num%i==0 && num!=1)
                {
                        flag=1;
         }
        }
        if(flag=1)
        printf("number is COMPOSITE number");
        else
        printf("number is prime number");
        if(num==1)
        printf("it is a unique number",num);
        else if(num==2)
        printf("it is a prime number",num);
}
```

OUTPUT:



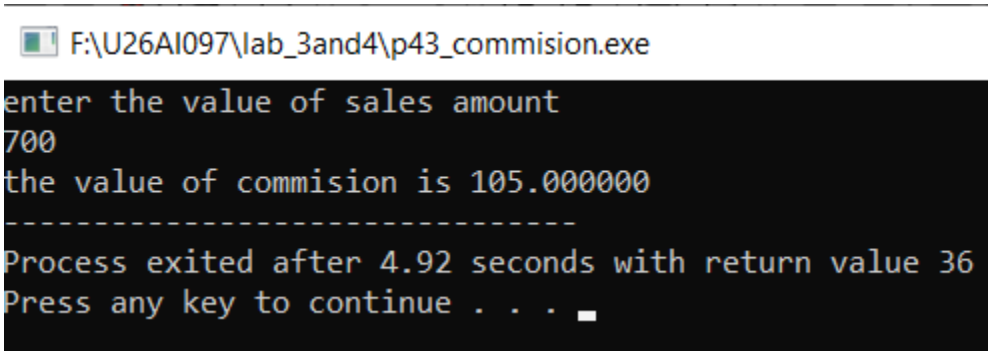
Q:43. WRITE A C PROGRAM TO CALCULATE THE COMMISSION FOR A SALES REPRESENTATIVE AS PER THE SALES AMOUNT GIVEN BELOW. If sales<=Rs. 500, commission is 5% If sales> but <=2000, commission is Rs. 35 plus 10% above Rs. 500 If sales>2000 but <=5000, commission is Rs. 185 plus 12% above Rs. 2000 If sales>5000, commission is 12.5%

CODE:

```
#include<stdio.h>
int main()
{
	float commision,amount;
	printf("enter the value of sales amount\n");
	scanf("%f",&amount);
	if(amount<=500)
	{
		commision=0.05*amount;
		printf("the value of commision is %f",commision);
	}
	else if(amount>500 && amount<=2000)
	{
		commision=0.1*amount+35;
		printf("the value of commision is %f",commision);
	}
	else if(amount>2000 && amount<=5000)
	{
		commision=0.12*amount+185;
		printf("the value of commision is %f",commision);
```

```
    }
    else if(amount>5000)
    {
        commision=0.125*amount;
        printf("the value of commision is %f",commision);
    }
    else
    printf("no commision");
}
```

OUTPUT:



Q:

44. WRITE A PROGRAM IN C FOR THE FOLLOWING.
AN ELECTRIC DISTRIBUTION COMPANY CHARGES ITS DOMESTIC CONSUMERS AS FOLLOWS.

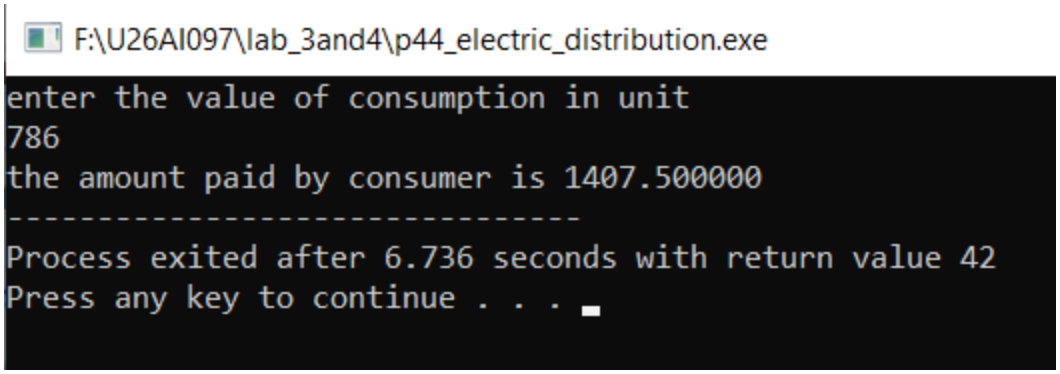
Consumption in unit	Rate for Charge
0-200	Re 0.50 per unit
210-400	Rs. 100 plus Re 0.65 per unit excess of 200
401-600	Rs. 230 plus Re 0.80 per unit excess of 400
Above 600	Rs. 425 plus Rs. 125 per unit excess of 600

Print the amount to be paid by the consumer.

CODE:

```
#include<stdio.h>
int main()
{
	float consumption,paidamt;
	printf("enter the value of consumption in unit\n");
	scanf("%f",&consumption);
	if(consumption>=0 && consumption<=200)
	{
		paidamt=0.50*consumption;
		printf("the amount paid by consumer is %f",paidamt);
	}
	else if(consumption>=210 && consumption<=400)
	{
		paidamt=0.65*consumption+100;
		printf("the amount paid by consumer is %f",paidamt);
	}
	else if(consumption>=401 && consumption<=600)
	{
		paidamt=0.80*consumption+230;
		printf("the amount paid by consumer is %f",paidamt);
	}
	else
	{
		paidamt=425+1.25*consumption;
		printf("the amount paid by consumer is %f",paidamt);
	}
}
```

OUTPUT:



LAB 5:-

36) A
AB
ABC
ABCD
ABCDE

Code:-

```
#include<stdio.h>
int main()
{
    int i,j;
    for(i=1;i<=5;i++)
    {
        for(j=1;j<=i;j++)
        {
            printf("%c",i+64);
        }
        printf("\n");
    }
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5> & "C:\Users\HET\Documents\c programming\U26AI097\lab 5\p36_abc_pattern.exe"
A
BB
CCC
DDDD
EEEEE
```

37)

```
    1
   12
  123
 1234
12345
```

Code:-

```c
#include<stdio.h>
int main()
{
    int i,j;
    for(i=1;i<=5;i++)
    {
        for(j=1;j<=5-i;j++)
        {
            printf(" ");
        }
        for(j=1;j<=i;j++)
        {
            printf("%d",j);
        }
        printf("\n");
    }
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5> & "C:\Users\HET\Documents\c programming\U26AI097\lab 5\p37_123.exe"
    1
   12
  123
 1234
12345
```

38)

```
*
  **
 ***
  ****
 *****
```

Code:-

```
#include<stdio.h>
int main()
{
    int i,j;
    for(i=1;i<=5;i++)
    {
        if(i%2!=0)
        {
        for(j=1;j<=i;j++)
        {
            printf("*");
        }
        printf("\n");
        }
        if(i%2==0)
        {
            printf(" ");
         for(j=2;j<=i+1;j++)
         {
            printf("*");
         }
         printf("\n");
        }
      }
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5> & "C:\Users\HET\Documents\c programming\U26AI097\lab 5\p38_star.exe"
*
 **
***
 ****
*****
```

39) A

```
  BB
 CCC
  DDDD
  EEEEE
```

Code:-

```c
#include<stdio.h>
int main()
{
    int i,j;
    for(i=1;i<=5;i++)
    {
        if(i%2!=0)
        {
        for(j=1;j<=i;j++)
        {
            printf("%c",i+64);
        }
        printf("\n");
        }
        if(i%2==0)
        {
            printf(" ");
         for(j=2;j<=i+1;j++)
         {
            printf("%c",i+64);
         }
         printf("\n");
        }
    }
```

```
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5> & "C:\Users\HET\Documents\c programming\U26AI097\lab 5\p39_abc.exe"
A
 BB
CCC
 DDDD
EEEEE
```

40) 10000
01000
00100
00010
00001

Code:-

```
#include<stdio.h>
int main()
{
    int i,j;
    for(i=1;i<=5;i++)
    {
        for(j=1;j<=5;j++)
        if(j==i)
        {
            printf("1");
        }
        else
        {
            printf("0");
        }
        printf("\n");
    }
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5> & "C:\Users\HET\Documents\c programming\U26AI097\lab 5\p40_01pattern.exe"
10000
01000
00100
00010
00001
```

41)
```
   1
  121
 12321
1234321
```

Code:-

```c
#include<stdio.h>
int main()
{
    int i,j;
    for(i=1;i<=4;i++)
    {
        for(j=1;j<=4-i;j++)
        {
            printf(" ");
        }
        for(j=1;j<=i;j++)
        {
            printf("%d",j);
        }
        for(j=1;j<=i-1;j++)
        {
            printf("%d",j);
        }
        printf("\n");
    }
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5> & "C:\Users\HET\Documents\c programming\U26AI097\lab 5\p41_palindrom_pattern.exe"
   1
  121
 12312
1234123
```

42)

```
   *
  ***
 *****
*******
 *****
  ***
   *
```

Code:-

```
#include<stdio.h>
int main()
{
    int i,j;
    for(i=1;i<=4;i++)
    {
        for(j=1;j<=4-i;j++)
        {
            printf(" ");
        }
        for(j=1;j<=i;j++)
        {
            printf("*");
        }
        for(j=1;j<=i-1;j++)
        {
            printf("*");
        }
        printf("\n");
    }
    for(i=1;i<=4;i++)
    {
        for(j=1;j<=i;j++)
        {
            printf(" ");
        }
        for(j=1;j<=4-i;j++)
```

```
        {
            printf("*");
        }
        for(j=1;j<=3-i;j++)
        {
            printf("*");
        }

        printf("\n");
    }
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5>
    *
   ***
  *****
 *******
  *****
   ***
    *
```

43) PROGRAM TO RED AND ARRAY OF 10 INTEGER AND PRINT SUM OF NUMBERS.

Code:-

```
#include<stdio.h>
int main()
{
    int arr[10],i;
    int sum=0;
    printf("enter 10 numbers\n");
    for(i=0;i<10;i++)
    {
        scanf("%d",&arr[i]);
    }
    for(i=0;i<10;i++)
```

```
    {
        sum=sum+arr[i];
    }
    printf("sum of entered number is %d",sum);
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5>
enter 10 numbers
2
5
6
4
7
2
1
4
3
3
sum of entered number is 37
```

44) PROGRAM TO READ AN ARRAY OF 10 INTEGER AND COUNT TOTAL NO OF ODD AND TOTAL NO. OF EVERY ELEMENT.

Code:-

```
#include<stdio.h>
int main()
{
    int arr[10],i,count=0,sum_odd=0;
    printf("enter 10 numbers\n");
    for(i=0;i<10;i++)
    {
        scanf("%d",&arr[i]);
    }
    for(i=0;i<10;i++)
    {
```

```
        if(arr[i]%2!=0)
        {
        count=count+1;
        sum_odd=sum_odd+arr[i];
        }
    }
    printf("the total odd numbers entered are %d",count);
    printf("sum of odd numbers:%d",sum_odd);
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5>
enter 10 numbers
2
1
5
9
7
4
32
6
4
5
```

45) PROGRAM TO READ AN ARRAY OF 10 INTEGER AND COUNT TOTAL NO. OF POSITIVE, NEGATIVE, AND ZERO ELEMENTS.

Code:-

```
#include<stdio.h>
int main()
{
    int arr[10],i,posi_count=0,neg_count=0,zero_count=0;
    printf("enter 10 numbers\n");
    for(i=0;i<10;i++)
    {
```

```
        scanf("%d",&arr[i]);
    }
    for(i=0;i<10;i++)
    {
        if(arr[i]>0)
        {
            posi_count=posi_count+1;
        }
        else if(arr[i]<0)
        {
            neg_count=neg_count+1;
        }
        if(arr[i]==0)
        {
            zero_count=zero_count+1;
        }
    }
    printf("total positive numbers:%d\n",posi_count);
    printf("total negitive numbers:%d\n",neg_count);
    printf("total zero:%d",zero_count);
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5>
enter 10 numbers
2
1
30
0
5
4
8
9
7
0
total positive numbers:8
total negitive numbers:0
total zero:2
```

46) PROGRAM TO READ N NOS AND FINDOUT MAX AND SECOND MAX.

Code:-

```c
#include<stdio.h>
#include<limits.h>
int main()
{
    int n,i;
    printf("enter the number which you are going to enter\n");
    scanf("%d",&n);
    printf("enter the numbers\n");
    int number[n];
    for(i=0;i<n;i++)
    {
        scanf("%d",&number[i]);
    }
    int max=INT_MIN;
    for(i=0;i<n;i++)
    {
        if(max<number[i])
        max=number[i];
    }
```

```
    int smax=INT_MIN;
    for(i=0;i<n;i++)
    {
        if( number[i]!=max&&smax<number[i])
        smax=number[i];
    }
    printf("the max number is %d\n",max);
    printf("the smax number is %d",smax);
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5>
enter the number which you are going to enter
5
enter the numbers
2
3
9
5
1
the max number is 9
the smax number is 5
```

47) PROGRAM TO READ TWO ARRAYS OF 10 INTEGERS AND STORE ADDATION OF THOSE ARRAYS INTO THIRD.

Code:-

```
#include<stdio.h>
int main()
{
    int arr1[10],arr2[10],arr3[10],i;
    printf("enter 10 numbers for array 1\n");
    for(i=0;i<10;i++)
    {
        scanf("%d",&arr1[i]);
    }
    printf("enter 10 numbers for array 2\n");
```

```c
    for(i=0;i<10;i++)
    {
        scanf("%d",&arr2[i]);
    }
    for(i=0;i<10;i++)
    {
        arr3[i]=arr1[i]+arr2[i];
    }
    printf("array 3 is:\n");
    for(i=0;i<10;i++)
    {
        printf("%d\t",arr3[i]);
    }
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5> & "C:\Users\HET\Documents\c programming\U26AI097\lab 5\p47_lab5.exe"
enter 10 numbers for array 1
2
3
1
5
2
3
6
4
7
8
enter 10 numbers for array 2
4
3
2
6
4
7
8
9
1
2
array 3 is:
6       6       3       11      6       10      14      13      8       10
```

48) PROGRAM TO READ TWO ARRAYS OF 10 INTEGERS AND SWAP VALUES THESE ARRAYS.

Code:-

```c
#include<stdio.h>
int main()
```

```c
{
    int arr1[10],arr2[10],arr3[10],i;
    printf("enter 10 numbers for array 1\n");
    for(i=0;i<10;i++)
    {
        scanf("%d",&arr1[i]);
    }
    printf("enter 10 numbers for array 2\n");
    for(i=0;i<10;i++)
    {
        scanf("%d",&arr2[i]);
    }
    for(i=0;i<10;i++)
    {
        arr3[i]=arr1[i];
        arr1[i]=arr2[i];
        arr2[i]=arr3[i];
    }
    printf("array 1 after swapping: ");
    for(i=0;i<10;i++)
    {
        printf("%d\t",arr1[i]);
    }
    printf("\n");
    printf("array 2 after swapping: ");
    for(i=0;i<10;i++)
    {
        printf("%d\t",arr2[i]);
    }
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5> & "C:\Users\HET\Documents\c programming\U26AI097\lab 5\p48_lab5.exe"
enter 10 numbers for array 1
2
3
2
3
2
3
2
3
2
3
enter 10 numbers for array 2
1
2
3
4
5
6
7
8
9
0
array 1 after swapping: 1       2       3       4       5       6       7       8       9       0
array 2 after swapping: 2       3       2       3       2       3       2       3       2       3
```

49) PROGRAM TO SORT (SELECTION SORT) AN ARRAY.

Code:-

```c
#include<stdio.h>
int main()
{
    int n,i,x,j,min_idx;
    printf("enter the number of elements in array\n");
    scanf("%d",&n);
    int arr[n];
    printf("enter the elements of array\n");
    for(i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    for(i=0;i<n-1;i++)
    {
        min_idx = i;
        for(j=i+1;j<n;j++)
        {
            if(arr[j]<arr[min_idx])
            {
                min_idx=j;
            }
        }
```

```
        if(min_idx != i)
        {
            x=arr[i];
            arr[i]=arr[min_idx];
            arr[min_idx]=x;
        }
    }

    printf("after sorting the elements:\n");
    for(i=0;i<n;i++)
    {
        printf("%d\t",arr[i]);
    }
    return 0;
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5>
097\lab 5\p49_lab5.exe"
enter the number of elements in array
5
enter the elements of array
2
9
5
1
7
after sorting the elements:
1       2       5       7       9
```

50) PROGRAM TO SORT (BUBBLE SORT) AN ARRAY.

Code:-

```
#include<stdio.h>
int main()
{
    int n,i,x,j;
    printf("enter the number of elements in array\n");
```

```
    scanf("%d",&n);
    int arr[n];
    printf("enter the elements of array\n");
    for(i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    for(i=0;i<n;i++)
    {
        for(j=0;j<n-i;j++)
        {
        if(arr[j]>arr[j+1])
        {
            x=arr[j];
            arr[j]=arr[j+1];
            arr[j+1]=x;
        }
        }
    }
    printf("after sorting the elements:\n");
    for(i=0;i<n;i++)
    {
        printf("%d\t",arr[i]);
    }
}
```

Output:-

```
PS C:\Users\HET\Documents\c programming\U26AI097\lab 5
enter the number of elements in array
5
enter the elements of array
4
2
6
9
5
after sorting the elements:
2       4       5       6       9
```